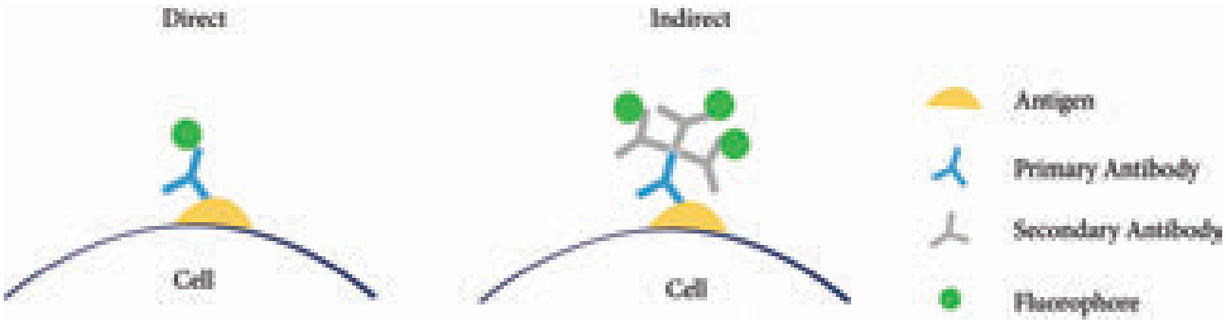


Figure 11.1: Methods in Immunofluorescence

determining site. Most of the antibodies have two binding sites and IgM has 5–10 binding sites.

Immunofluorescence

When antibodies are mixed with the fluorescent dyes such as fluorescein or rhodamine, they emit radiation. This phenomenon of emitting radiation by antibodies labelled with fluorescent dye is called immuno fluorescence. This reaction is well observed under fluorescent microscope. It is used to locate and identify antigens in tissues.

Types of Immunofluorescence

- Direct method
- Indirect method
- Sandwich method

Direct Method

In this method, the antibody labelled with fluorescent dye is directly applied on the tissue section. The labelled antibody binds with specific antigen. This can be observed under the fluorescent microscope.

Indirect Method

In this method, unlabelled antibodies are directly applied on the tissue sections which bind with the specific antigens. Then the antibody labelled with the fluorescent dye is added to the tissue. Anti-antibody specifically binds with already added or linked unlabelled antibody (Figure 11.1).

Sandwich Method

This is an immuno fluorescence method used to test the number of cells producing antibodies for a specific antigen. In this method, lymphocytes are fixed with ethanol. These fixed cells are treated with polysaccharide antigen of *Pneumococcus*. This antigen combines with those lymphocytes which have the capacity to produce antibody against pneumococcal antigen. Now fluorescent antibody is added. Antigen is sandwiched between antibodies.

ELISA (Enzyme Linked Immuno Sorbent Assay)

ELISA (Enzyme-Linked Immuno Sorbent Assay) is a plate-based assay technique designed for detecting and quantifying substances such as peptides, proteins, antibodies and hormones. It is also known as Enzyme Immuno Assay (EIA).

In 1971, after the descriptions of Peter Perlmann and Eva Engvall at Stockholm University in Sweden, ELISA has become the system of choice when assaying

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